SISTEMA FV CONECTADO A RED

PVGISAMPLIADOMULTI

Mi rrrr Real

P. Román

18 de June de 2026

Organización
Instalaciones S..
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Ayto. Villanueva
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Dirección: Plaza Mayor, 1

Resumen

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem.

pvgisampliadomulti

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Glosario de símbolos / nomenclatura

Variable	Descripción	Unidad
E_d	Producción de energía diaria promedio del sistema	kWh/d
E_m	Producción de energía mensual promedio del sistema	kWh/mo
E_y	Producción de energía anual promedio del sistema	kWh/y
H_d	Irradiación global diaria promedio por m^2 sobre los módulos	kWh/m2/d
H_m	Irradiación global mensual promedio por m^2 sobre los módulos	kWh/m2/mo
H_y	Irradiación global anual promedio por m^2 sobre los módulos	kWh/m2/y
SD_m	Desviación estándar de la producción mensual (variación interanual)	kWh
SD_y	Desviación estándar de la producción anual (variación interanual)	kWh
l_aoi	Pérdida por ángulo de incidencia	%
l_spec	Pérdida espectral	%
l_tg	Pérdida por temperatura e irradiancia	%
l_total	Pérdida Total del sistema	%
LCOE_pv	Costo nivelado de la electricidad fotovoltaica	moneda/kWh
System Cost	Costo Total de inversión del sistema	moneda
Interest	Tasa de interés anual aplicada al capital	%/y
Lifetime	Vida útil estimada del sistema fotovoltaico	años

Tabla 1: Glosario de símbolos / nomenclatura

1. Grupo FV G2.

Figura 1: (G2) Paneles FV.

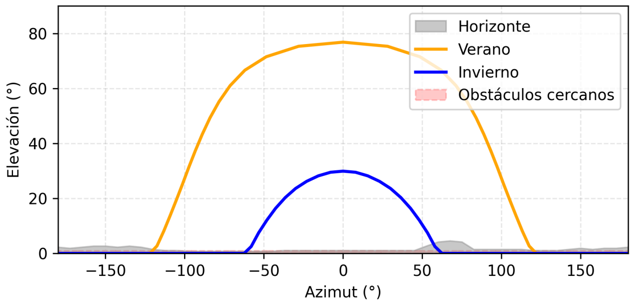

Figura 2: (G2) Perfil del horizonte y sombras.

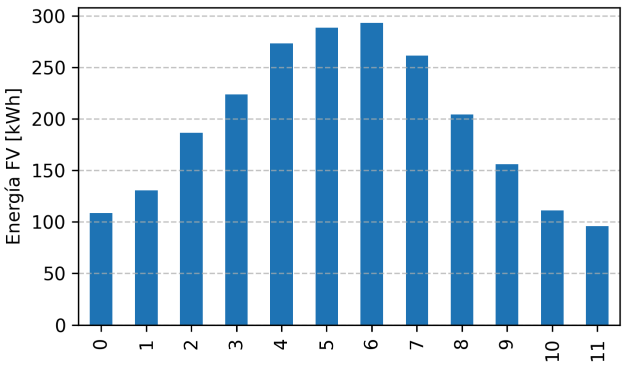

Figura 3: (G2) Producción de energía mensual.

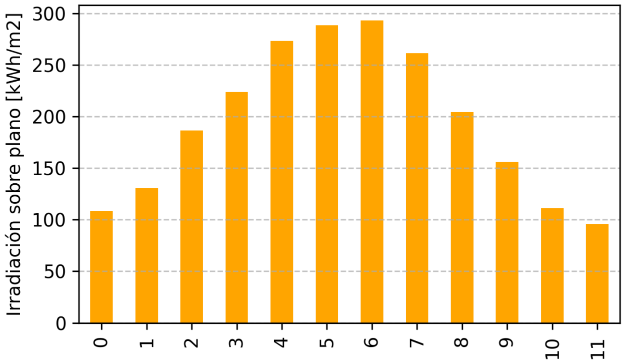

Figura 4: (G2) Irradiación mensual sobre plano fijo.

Parámetro	**Valor**
Latitud/Longitud	36.666, −4.462
Horizonte	DEM-calculated
Base de datos	PVGIS-SARAH3
Tecnología FV	c-Si
FV instalado	1.62 kWp
Pérdidas sistema	14.0%
Costo sistema	1620.0 €
Interés anual	5.0%
Vida útil	20 años

Tabla 2: (G2) Datos proporcionados

Parámetro	Valor
Ángulo de inclinación	0°
Ángulo de azimut	
Producción diaria (E_d)	6.39 kWh
Producción mensual (E_m)	194.42 kWh
Producción anual (E_y)	2332.98 kWh
Irradiación diaria (H_d)	5.15 kWh/m²
Irradiación mensual (H_m)	156.72 kWh/m²
Irradiación anual (H_y)	1880.65 kWh/m²
Desv. estándar mensual	3.8 kWh
Desv. estándar anual	45.63 kWh
Pérdida ángulo incidencia	−3.51%
Pérdida espectral	?(0)
Pérdida temp. e irradiancia	−7.72%
Pérdidas totales	−23.42%
LCOE	0.07 €/kWh

Tabla 3: (G2) Resultados de la simulación

month	E_d	E_m	SD_m
1	3.5	108.62	9.75
2	4.67	130.79	12.36
3	6.01	186.27	21.46
4	7.46	223.68	16.98
5	8.81	273.17	18.44
6	9.62	288.59	9.45
7	9.46	293.26	6.06
8	8.43	261.23	7.69
9	6.81	204.27	8.93
10	5.03	156.0	11.35
11	3.71	111.24	9.59
12	3.09	95.88	7.74

Tabla 4: (G2) Energía FV mensual

2. Grupo FV G3.

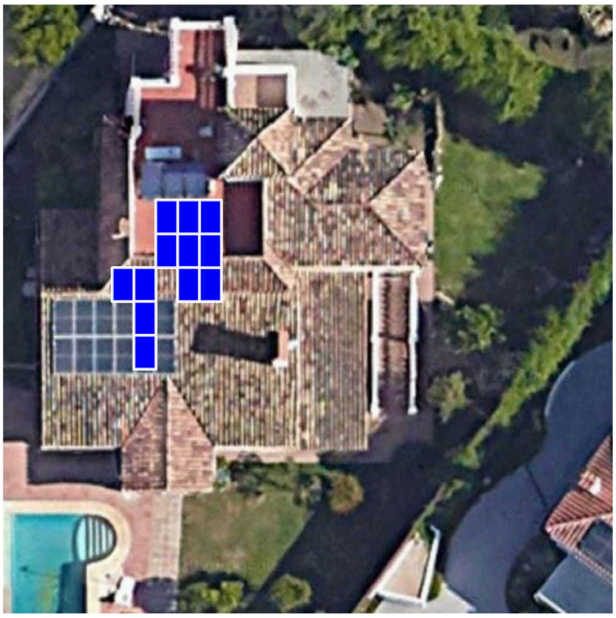

Figura 5: (G3) Paneles FV.

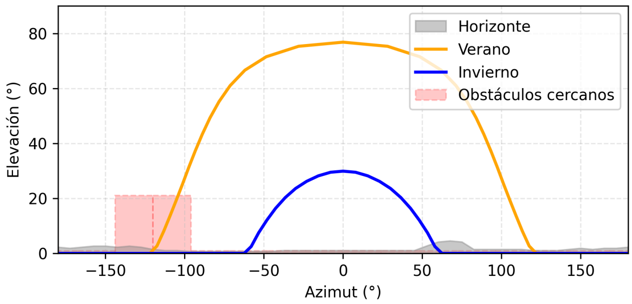

Figura 6: (G3) Perfil del horizonte y sombras.

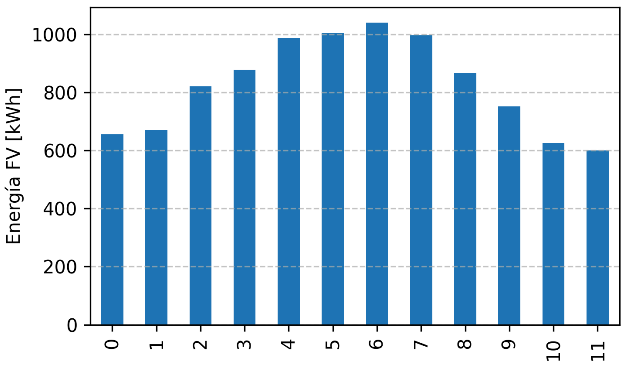

Figura 7: (G3) Producción de energía mensual.

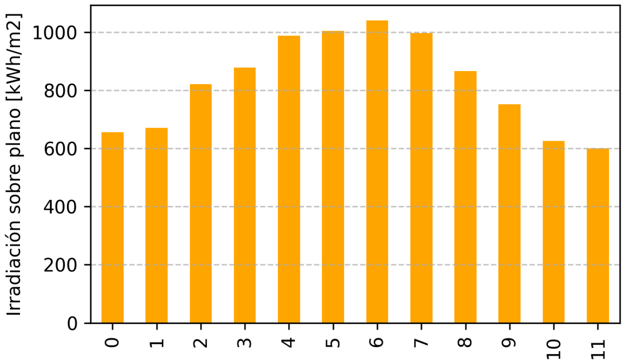

Figura 8: (G3) Irradiación mensual sobre plano fijo.

Parámetro	**Valor**
Latitud/Longitud	36.666, −4.462
Horizonte	DEM-calculated
Base de datos	PVGIS-SARAH3
Tecnología FV	c-Si
FV instalado	6.0 kWp
Pérdidas sistema	14.0%
Costo sistema	5994.0 €
Interés anual	5.0%
Vida útil	20 años

Tabla 5: (G3) Datos proporcionados

Parámetro	Valor
Ángulo de inclinación	26°
Ángulo de azimut	0°
Producción diaria (E_d)	27.12 kWh
Producción mensual (E_m)	825.0 kWh
Producción anual (E_y)	9899.99 kWh
Irradiación diaria (H_d)	5.86 kWh/m²
Irradiación mensual (H_m)	178.33 kWh/m²
Irradiación anual (H_y)	2139.94 kWh/m²
Desv. estándar mensual	20.73 kWh
Desv. estándar anual	248.81 kWh
Pérdida ángulo incidencia	−2.65%
Pérdida espectral	?(0)
Pérdida temp. e irradiancia	−7.9%
Pérdidas totales	−22.9%
LCOE	0.06 €/kWh

Tabla 6: (G3) Resultados de la simulación

month	E_d	E_m	SD_m
1	21.16	655.88	70.97
2	23.97	671.19	77.24
3	26.48	820.85	108.18
4	29.26	877.71	72.07
5	31.86	987.76	67.16
6	33.48	1004.48	32.69
7	33.57	1040.59	21.18
8	32.15	996.5	30.23
9	28.88	866.35	43.8
10	24.27	752.44	66.87
11	20.86	625.83	65.09
12	19.37	600.41	60.4

Tabla 7: (G3) Energía FV mensual

3. Grupo FV Total.

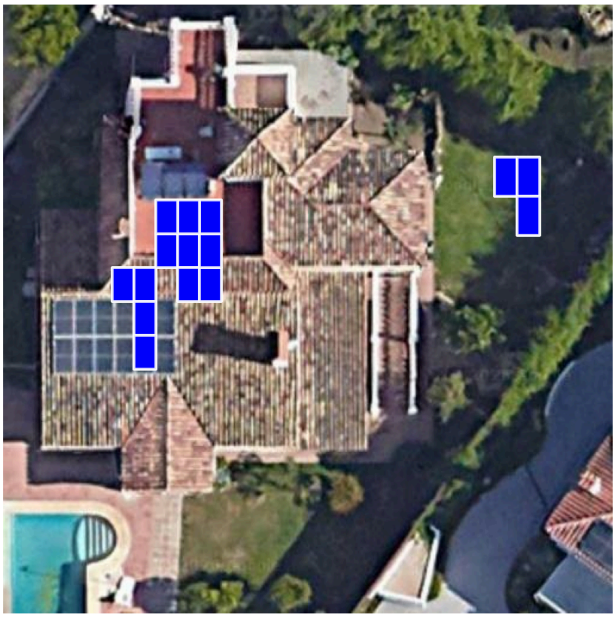

Figura 9: (Total) Paneles FV.

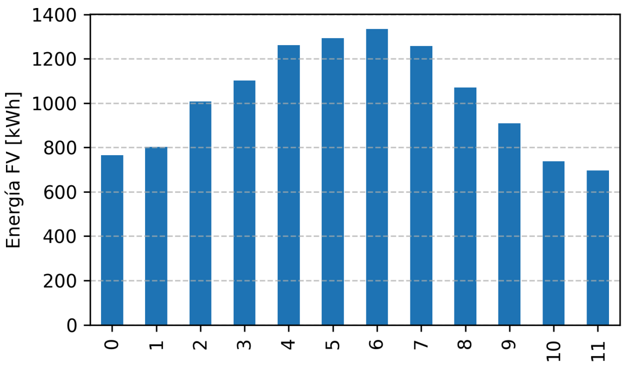

Figura 10: (Total) Producción de energía mensual.

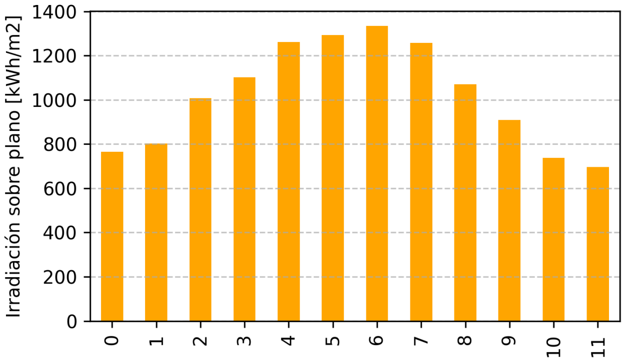

Figura 11: (Total) Irradiación mensual sobre plano fijo.

Parámetro	**Valor**
Latitud/Longitud	36.67, −4.46
Horizonte	DEM-calculated
Base de datos	PVGIS-SARAH3
Tecnología FV	c-Si
FV instalado	7.62 kWp
Pérdidas sistema	14.0%
Costo sistema	1620.0 €
Interés anual	5.0%
Vida útil	20 años

Tabla 8: (Total) Datos proporcionados

Parámetro	**Valor**
Ángulo de inclinación	0°
Ángulo de azimut	
Producción diaria (E_d)	33.51 kWh
Producción mensual (E_m)	1019.42 kWh
Producción anual (E_y)	12232.97 kWh
Irradiación diaria (H_d)	5.71 kWh/m²
Irradiación mensual (H_m)	173.74 kWh/m²
Irradiación anual (H_y)	2084.82 kWh/m²
Desv. estándar mensual	24.53 kWh
Desv. estándar anual	294.44 kWh
Pérdida ángulo incidencia	−2.83%
Pérdida espectral	0.0
Pérdida temp. e irradiancia	−7.86%
Pérdidas totales	−23.01%
LCOE	0.06 €/kWh

Tabla 9: (Total) Resultados de la simulación

month	E_d	E_m	SD_m
1	24.66	764.5	80.72
2	28.64	801.98	89.6
3	32.49	1007.12	129.64
4	36.72	1101.39	89.05
5	40.67	1260.93	85.6
6	43.1	1293.07	42.14
7	43.03	1333.85	27.24
8	40.58	1257.73	37.92
9	35.69	1070.62	52.73
10	29.3	908.44	78.22
11	24.57	737.07	74.68
12	22.46	696.29	68.14

Tabla 10: (Total) Energía FV mensual

4. Presupuesto

Capítulo	Categoría	Variable	Fijo (€)	€/Wdc	MO (%)	Mat	MO	Total
Coste Directo	Equipos Principales	Paneles fotovoltaicos monocristalinos	0.0	0.2	5.0	0.2	0.0	0.2
		Inversor string/híbrido	150.0	0.1	5.0	142.6	7.5	150.1
		Sistema de almacenamiento (Batería)	3000.0	0.2	5.0	2850.1	150.0	3000.1
	Balance de Sistema (BOS)	Soportes, cableado y protecciones	750.0	0.2	40.0	450.1	300.1	750.2
		Sistema de monitorización	150.0	0.0	10.0	135.0	15.0	150.0
	Obra Civil	Adecuación y pequeñas adaptaciones	200.0	0.0	80.0	40.0	160.0	200.0
Coste Directo - Subtotal			4250.0	0.7	145.0	3618.0	632.6	4250.6
Coste Indirecto	Ingeniería	Proyecto, dirección y legalización	900.0	0.0	100.0	0.0	900.0	900.0
	Interconexión	Trámites con distribuidora y tasas	500.0	0.0	90.0	50.0	450.0	500.0
	Gastos Generales	Margen comercial y overhead	0.0	0.2	0.0	0.2	0.0	0.2
	Contingencias	Reserva para imprevistos (3-5%)	0.0	0.0	0.0	0.0	0.0	0.0
Coste Indirecto - Subtotal			1400.0	0.2	190.0	50.2	1350.0	1400.2
TOTAL GENERAL			5650.0	1.0		3668.2	1982.6	5650.8

Tabla 11: Presupuesto

5. inputs

Categoría	Variable	valor
Proyecto	nombre	Mi rrrr Real
Autor		P. Román
	N Registro	N.12345
Organización	nombre	Instalaciones S..
	CIF	A12345678
	dirección	Calle Iglesia, 123
	teléfono	912 34 56 78
	correo	pperezinstalaciones.com
Destinatario	nombre	Ayto. Villanueva
	CIF	
	dirección	Plaza Mayor, 1

Tabla 12: PDF

Capítulo	Categoría	nombre	tipoEstructura	id	lat	lng	rotation	width	height	tilt	slope	orientation	modelo	potenciaW	rows	cols	FVcost	paneles	sombras	n_paneles	peak_power	cost
FV	Paneles	G2	coplanar	178177821236466624290549491510030795912				1.134	2.278	30	0	vertical	JKM540M-72HL4-V	540	1	1	1000	[“0,0”,“0,1”,“1,1”][1,1,1,1,1,1,31,1,1,1,1,1,1,1,1,62]				1620.0
		G3	coplanar	178099716823566624290549491510030795912				1.1	1.9	26	26	vertical		500	1	1	999	[“1,-2”,“1,-1”[“2,1-2”1,“3,1-1”,1020,1”,“0,1,1”,“1,6,1”,“1,2”],“0,2”,“5991420”,“−1,1”,“−				

Tabla 13: Diseño FV

Capítulo	Categoría	Subcategoría	Consumo_diario_kWh	Perfil_24h
Residencial	Vivienda Estándar	Uso General	10	[0.024, 0.016, 0.016, 0.016, 0.016, 0.024, 0.032, 0.145, 0.04, 0.032, 0.024, 0.024, 0.032, 0.04, 0.032, 0.024, 0.032, 0.047, 0.063, 0.08, 0.095, 0.071, 0.047, 0.032]
	Vivienda Unifamiliar	Aire Acondicionado	12	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.02, 0.05, 0.08, 0.12, 0.15, 0.15, 0.12, 0.1, 0.08, 0.06, 0.04, 0.02, 0.01, 0.0, 0.0, 0.0]
		Calefacción (B. Calor)	20	[0.05, 0.06, 0.07, 0.07, 0.07, 0.08, 0.1, 0.12, 0.08, 0.05, 0.03, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.03, 0.04, 0.05, 0.06, 0.06, 0.04, 0.04]
		Cargador Coche Eléctrico	15	[0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.04, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.04]
	Bloque Pisos	Servicios Comunes	30	[0.0416, 0.0416]
Terciario	Oficina Moderna	Iluminación LED	15	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
		Climatización	50	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.12, 0.12, 0.12, 0.12, 0.05, 0.12, 0.12, 0.06, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
	Comercio	Rótulos Luminosos	5	[0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1]
Industrial	Nave Producción	Maquinaria Pesada	350	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.15, 0.15, 0.15, 0.15, 0.15, 0.05, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
		Aire Comprimido	40	[0.0416, 0.0416]
	Almacén Logístico	Carga Carretillas	60	[0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.1, 0.1, 0.1]
Agrario	Explotación Agrícola	Bombeo Riego	150	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.15, 0.15, 0.15, 0.15, 0.15, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
	Granja Ganadera	Ventilación Forzada	25	[0.0416, 0.0416]
		Ordeño Automático	15	[0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0]

Tabla 14: CONSUMO

Capítulo	Categoría	Variable	Fijo (€)	€/Wdc	MO (%)
Coste Directo	Equipos Principales	Paneles fotovoltaicos monocristalinos	0	0.22	5
		Inversor string/híbrido	150	0.12	5
		Sistema de almacenamiento (Batería)	3000	0.15	5
	Balance de Sistema (BOS)	Soportes, cableado y protecciones	750	0.19	40
		Sistema de monitorización	150	0.02	10
	Obra Civil	Adecuación y pequeñas adaptaciones	200	0.005	80
Coste Indirecto	Ingeniería	Proyecto, dirección y legalización	900	0.02	100
	Interconexión	Trámites con distribuidora y tasas	500	0.01	90
	Gastos Generales	Margen comercial y overhead	0	0.18	0
	Contingencias	Reserva para imprevistos (3-5%)	0	0.04	0
Otros	Otros		0	0.0	0

Tabla 15: COSTES DE INSTALACION

Capítulo	Categoría	Concepto	Año_inicio	Año_fin	Fijo (€/año)	Variable	Unidad_variable
Operación FV	Energía fotovoltaica	PV_Fijo_Anual	1	25	0.0	0	€/año
		PV_Fijo_por_Capacidad	1	25	19.0	0	€/Wdc
		PV_Variable_Generacion	1	25	0.0	0	€/kWh
Batería	Almacenamiento	Bateria_Fijo_Anual	1	15	0.0	0	€/año
		Bateria_Fijo_por_Capacidad	1	15	7.25	0	€/kWh_instalada
		Bateria_Variable_Descarga	1	15	0.0	0	€/kWh_descargado
	Reemplazo de activos	Bateria_Coste_Reemplazo	10	10	252.0	0	€/kWh_instalada
Infraestructura	Uso de suelo	Arrendamiento_Anual_Terreno	1	25	0.0	0	€/año
Otrsos	Otdros	Otros	1	25	0.0	0	€

Tabla 16: COSTES DE OPERACION

Capítulo	Categoría	Concepto	Año_inicio	Año_fin	Ingreso_fijo (€/año)	Ingreso_variable	Unidad_variable
Ingresos de explotación	Autoconsumo	Ahorro autoconsumo	1	25	0	0.18	€/kWh autoconsumido
	Excedentes	Venta excedentes	1	25	0	0.08	€/kWh exportado
		Compensación simplificada	1	25	0	0.06	€/kWh compensado
Ingresos regulados	Certificados	CAE	1	10	0	0.14	€/kWh generado
		Garantías de origen	1	25	0	0.005	€/kWh generado
Subvenciones	Bonificaciones fiscales	Bonificación IBI	1	5	100	0.0	€/año
	Deducciones fiscales	Deducción IRPF	1	1	200	0.0	€/año
	Inversión	Next Generation	1	1	0	0.3	% CAPEX
	Locales/autonómicas	ICIO	1	1	100	0.0	€/año
Otros	Otros	Otros	1	25	0	0.0	€

Tabla 17: INGRESOS

Capítulo	**Categoría**	**Parámetro**	**Valor**	**Unidad**
Análisis Financiero	Proyecto	Duración de la actuación	2033.0	años
		Impuestos (IVA)	21.0	%
	Mercado	Tasa de inflación	2.5	%/año
		Precio de la energía	0.12	€/kWh
		Tasa de descuento (VAN)	0.08	-
	Financiación	Porcentaje financiado	27.0	%
		Años de financiación	10.0	años
		Tipo de interés anual	7.0	%
	Operación	Tasa seguro anual	0.0	% de instalación

Tabla 18: FINANCIEROS

6. printhorizon

6.1. printhorizon

- **inputs**:
  - **location**:
    - **latitude**: 36.66624290569493
    - **longitude**: −4.461510030795912
    - **elevation**: 4.0
  - **horizon_db**: DEM-calculated
- **outputs**:
  - **horizon_profile**:
    - **A**: −180.0
    - **H_hor**: 2.3
    - **A**: −172.5
    - **H_hor**: 1.9
    - **A**: −165.0
    - **H_hor**: 2.3
    - **A**: −157.5
    - **H_hor**: 2.7
    - **A**: −150.0
    - **H_hor**: 2.7
    - **A**: −142.5
    - **H_hor**: 2.3
    - **A**: −135.0
    - **H_hor**: 2.7
    - **A**: −127.5
    - **H_hor**: 2.3
    - **A**: −120.0
    - **H_hor**: 1.5
    - **A**: −112.5
    - **H_hor**: 1.1
    - **A**: −105.0
    - **H_hor**: 1.1
    - **A**: −97.5
    - **H_hor**: 0.8
    - **A**: −90.0
    - **H_hor**: 0.8
    - **A**: −82.5
    - **H_hor**: 0.8
    - **A**: −75.0
    - **H_hor**: 0.8
    - **A**: −67.5
    - **H_hor**: 0.8
    - **A**: −60.0
    - **H_hor**: 0.8
    - **A**: −52.5
    - **H_hor**: 0.8
    - **A**: −45.0
    - **H_hor**: 0.8
    - **A**: −37.5
    - **H_hor**: 1.1
    - **A**: −30.0
    - **H_hor**: 1.1
    - **A**: −22.5
    - **H_hor**: 1.1
    - **A**: −15.0
    - **H_hor**: 1.1
    - **A**: −7.5
    - **H_hor**: 1.1
    - **A**: 0.0
    - **H_hor**: 1.1
    - **A**: 7.5
    - **H_hor**: 1.1
    - **A**: 15.0
    - **H_hor**: 1.1
    - **A**: 22.5
    - **H_hor**: 1.1
    - **A**: 30.0
    - **H_hor**: 1.1
    - **A**: 37.5
    - **H_hor**: 1.1
    - **A**: 45.0
    - **H_hor**: 1.1
    - **A**: 52.5
    - **H_hor**: 2.7
    - **A**: 60.0
    - **H_hor**: 4.2
    - **A**: 67.5
    - **H_hor**: 4.6
    - **A**: 75.0
    - **H_hor**: 4.2
    - **A**: 82.5
    - **H_hor**: 1.5
    - **A**: 90.0
    - **H_hor**: 1.5
    - **A**: 97.5
    - **H_hor**: 1.5
    - **A**: 105.0
    - **H_hor**: 1.5
    - **A**: 112.5
    - **H_hor**: 1.5
    - **A**: 120.0
    - **H_hor**: 1.1
    - **A**: 127.5
    - **H_hor**: 1.1
    - **A**: 135.0
    - **H_hor**: 1.1
    - **A**: 142.5
    - **H_hor**: 1.5
    - **A**: 150.0
    - **H_hor**: 1.9
    - **A**: 157.5
    - **H_hor**: 1.5
    - **A**: 165.0
    - **H_hor**: 1.9
    - **A**: 172.5
    - **H_hor**: 1.9
    - **A**: 180.0
    - **H_hor**: 2.3
  - **winter_solstice**:
    - **A_sun(w)**: −180.0
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −151.9
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −131.5
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −118.0
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −108.7
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −101.7
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −96.0
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −91.1
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −86.7
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −82.6
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −78.7
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −74.8
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −70.8
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −66.8
    - **H_sun(w)**: 0.0
    - **A_sun(w)**: −62.5

- **H_sun(w)**: 0.0
- **A_sun(w)**: −58.0
- **H_sun(w)**: 2.5
- **A_sun(w)**: −53.3
- **H_sun(w)**: 7.5
- **A_sun(w)**: −48.1
- **H_sun(w)**: 12.1
- **A_sun(w)**: −42.6
- **H_sun(w)**: 16.4
- **A_sun(w)**: −36.5
- **H_sun(w)**: 20.3
- **A_sun(w)**: −30.0
- **H_sun(w)**: 23.6
- **A_sun(w)**: −23.0
- **H_sun(w)**: 26.2
- **A_sun(w)**: −15.6
- **H_sun(w)**: 28.2
- **A_sun(w)**: −7.9
- **H_sun(w)**: 29.5
- **A_sun(w)**: 0.0
- **H_sun(w)**: 29.9
- **A_sun(w)**: 7.9
- **H_sun(w)**: 29.5
- **A_sun(w)**: 15.6
- **H_sun(w)**: 28.2
- **A_sun(w)**: 23.0
- **H_sun(w)**: 26.2
- **A_sun(w)**: 30.0
- **H_sun(w)**: 23.6
- **A_sun(w)**: 36.5
- **H_sun(w)**: 20.3
- **A_sun(w)**: 42.6
- **H_sun(w)**: 16.4
- **A_sun(w)**: 48.1
- **H_sun(w)**: 12.1
- **A_sun(w)**: 53.3
- **H_sun(w)**: 7.5
- **A_sun(w)**: 58.0
- **H_sun(w)**: 2.5
- **A_sun(w)**: 62.5
- **H_sun(w)**: 0.0
- **A_sun(w)**: 66.8
- **H_sun(w)**: 0.0
- **A_sun(w)**: 70.8
- **H_sun(w)**: 0.0
- **A_sun(w)**: 74.8
- **H_sun(w)**: 0.0
- **A_sun(w)**: 78.7
- **H_sun(w)**: 0.0
- **A_sun(w)**: 82.6
- **H_sun(w)**: 0.0
- **A_sun(w)**: 86.7
- **H_sun(w)**: 0.0
- **A_sun(w)**: 91.1
- **H_sun(w)**: 0.0
- **A_sun(w)**: 96.0
- **H_sun(w)**: 0.0
- **A_sun(w)**: 101.7
- **H_sun(w)**: 0.0
- **A_sun(w)**: 108.7
- **H_sun(w)**: 0.0
- **A_sun(w)**: 118.0
- **H_sun(w)**: 0.0
- **A_sun(w)**: 131.5
- **H_sun(w)**: 0.0
- **A_sun(w)**: 151.9
- **H_sun(w)**: 0.0
- **A_sun(w)**: 180.0
- **H_sun(w)**: 0.0

- **summer_solstice**:
  - **A_sun(s)**: −180.0
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −172.1
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −164.4
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −157.0
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −150.0
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −143.5
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −137.4
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −131.9
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −126.7
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −122.0
  - **H_sun(s)**: 0.0
  - **A_sun(s)**: −117.5
  - **H_sun(s)**: 2.7
  - **A_sun(s)**: −113.2
  - **H_sun(s)**: 8.1
  - **A_sun(s)**: −109.2
  - **H_sun(s)**: 13.7
  - **A_sun(s)**: −105.2
  - **H_sun(s)**: 19.5
  - **A_sun(s)**: −101.3
  - **H_sun(s)**: 25.3
  - **A_sun(s)**: −97.4
  - **H_sun(s)**: 31.3
  - **A_sun(s)**: −93.3
  - **H_sun(s)**: 37.3
  - **A_sun(s)**: −88.9
  - **H_sun(s)**: 43.3
  - **A_sun(s)**: −84.0
  - **H_sun(s)**: 49.3
  - **A_sun(s)**: −78.3
  - **H_sun(s)**: 55.2
  - **A_sun(s)**: −71.3
  - **H_sun(s)**: 61.0
  - **A_sun(s)**: −62.0
  - **H_sun(s)**: 66.6
  - **A_sun(s)**: −48.5
  - **H_sun(s)**: 71.5
  - **A_sun(s)**: −28.1
  - **H_sun(s)**: 75.3
  - **A_sun(s)**: 0.0
  - **H_sun(s)**: 76.8
  - **A_sun(s)**: 28.1
  - **H_sun(s)**: 75.3
  - **A_sun(s)**: 48.5
  - **H_sun(s)**: 71.5
  - **A_sun(s)**: 62.0
  - **H_sun(s)**: 66.6
  - **A_sun(s)**: 71.3
  - **H_sun(s)**: 61.0
  - **A_sun(s)**: 78.3
  - **H_sun(s)**: 55.2
  - **A_sun(s)**: 84.0
  - **H_sun(s)**: 49.3
  - **A_sun(s)**: 88.9
  - **H_sun(s)**: 43.3
  - **A_sun(s)**: 93.3
  - **H_sun(s)**: 37.3
  - **A_sun(s)**: 97.4
  - **H_sun(s)**: 31.3
  - **A_sun(s)**: 101.3
  - **H_sun(s)**: 25.3
  - **A_sun(s)**: 105.2

    - **H_sun(s)**: 19.5
    - **A_sun(s)**: 109.2
    - **H_sun(s)**: 13.7
    - **A_sun(s)**: 113.2
    - **H_sun(s)**: 8.1
    - **A_sun(s)**: 117.5
    - **H_sun(s)**: 2.7
    - **A_sun(s)**: 122.0
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 126.7
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 131.9
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 137.4
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 143.5
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 150.0
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 157.0
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 164.4
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 172.1
    - **H_sun(s)**: 0.0
    - **A_sun(s)**: 180.0
    - **H_sun(s)**: 0.0
- **meta**:
  - **inputs**:
    - **location**:
      - **description**: Selected location
      - **variables**:
        - **latitude**:
          - **description**: Latitude
          - **units**: decimal degree
        - **longitude**:
          - **description**: Longitude
          - **units**: decimal degree
        - **elevation**:
          - **description**: Elevation
          - **units**: m
    - **horizon_db**:
      - **description**: Source of horizon data
  - **outputs**:
    - **horizon_profile**:
      - **type**: series
      - **description**: Horizon profile
      - **variables**:
        - **A**:
          - **description**: Azimuth (0 = S, 90 = W, −90 = E)
          - **units**: degree
        - **H_hor**:
          - **description**: Horizon height
          - **units**: degree
    - **winter_solstice**:
      - **type**: series
      - **description**: Winter solstice
      - **variables**:
        - **A_sun(w)**:
          - **description**: Sun azimuth in the winter solstice (Dec 21) (0 = S, 90 = W, −90 = E)
          - **units**: degree
        - **H_sun(w)**:
          - **description**: Sun height in the winter solstice (Dec 21)
          - **units**: degree
    - **summer_solstice**:
      - **type**: series
      - **description**: Summer solstice profile
      - **variables**:
        - **A_sun(s)**:
          - **description**: Sun azimuth in the summer solstice (June 21) (0 = S, 90 = W, −90 = E)
          - **units**: degree
        - **H_sun(s)**:
          - **description**: Sun height in the summer solstice (June 21)
          - **units**: degree

7. printhorizon

7.1. G2

- **inputs**:
  - **location**:
    - **latitude**: 36.66624290569493
    - **longitude**: −4.461510030795912
    - **elevation**: 4.0
  - **meteo_data**:
    - **radiation_db**: PVGIS-SARAH3
    - **meteo_db**: ERA5
    - **year_min**: 2005
    - **year_max**: 2023
    - **use_horizon**: True
    - **horizon_db**: DEM-calculated
  - **mounting_system**:
    - **fixed**:
      - **slope**:
        - **value**: 0
        - **optimal**: False
      - **azimuth**:
        - **value**: 0
        - **optimal**: False
      - **type**: free-standing
  - **pv_module**:
    - **technology**: c-Si
    - **peak_power**: 1.62
    - **system_loss**: 14.0
  - **economic_data**:
    - **system_cost**: 1620.0
    - **interest**: 5.0
    - **lifetime**: 20
- **outputs**:
  - **monthly**:
    - **fixed**:
      - **month**: 1
      - **E_d**: 3.5
      - **E_m**: 108.62
      - **H(i)_d**: 2.78
      - **H(i)_m**: 86.25
      - **SD_m**: 9.75
      - **month**: 2
      - **E_d**: 4.67
      - **E_m**: 130.79
      - **H(i)_d**: 3.63
      - **H(i)_m**: 101.69
      - **SD_m**: 12.36
      - **month**: 3
      - **E_d**: 6.01
      - **E_m**: 186.27
      - **H(i)_d**: 4.68
      - **H(i)_m**: 145.12
      - **SD_m**: 21.46
      - **month**: 4
      - **E_d**: 7.46
      - **E_m**: 223.68
      - **H(i)_d**: 5.87
      - **H(i)_m**: 176.25
      - **SD_m**: 16.98
      - **month**: 5
      - **E_d**: 8.81
      - **E_m**: 273.17
      - **H(i)_d**: 7.06
      - **H(i)_m**: 218.93
      - **SD_m**: 18.44
      - **month**: 6
      - **E_d**: 9.62
      - **E_m**: 288.59
      - **H(i)_d**: 7.85
      - **H(i)_m**: 235.42
      - **SD_m**: 9.45
      - **month**: 7
      - **E_d**: 9.46
      - **E_m**: 293.26
      - **H(i)_d**: 7.84
      - **H(i)_m**: 243.17
      - **SD_m**: 6.06
      - **month**: 8
      - **E_d**: 8.43
      - **E_m**: 261.23
      - **H(i)_d**: 6.97
      - **H(i)_m**: 216.12
      - **SD_m**: 7.69
      - **month**: 9
      - **E_d**: 6.81
      - **E_m**: 204.27
      - **H(i)_d**: 5.54
      - **H(i)_m**: 166.16
      - **SD_m**: 8.93
      - **month**: 10
      - **E_d**: 5.03
      - **E_m**: 156.0
      - **H(i)_d**: 4.06
      - **H(i)_m**: 125.83
      - **SD_m**: 11.35
      - **month**: 11
      - **E_d**: 3.71
      - **E_m**: 111.24
      - **H(i)_d**: 2.96
      - **H(i)_m**: 88.76
      - **SD_m**: 9.59
      - **month**: 12
      - **E_d**: 3.09
      - **E_m**: 95.88
      - **H(i)_d**: 2.48
      - **H(i)_m**: 76.97
      - **SD_m**: 7.74
  - **totals**:
    - **fixed**:
      - **E_d**: 6.39
      - **E_m**: 194.42
      - **E_y**: 2332.98
      - **H(i)_d**: 5.15
      - **H(i)_m**: 156.72
      - **H(i)_y**: 1880.65
      - **SD_m**: 3.8
      - **SD_y**: 45.63
      - **l_aoi**: −3.51
      - **l_spec**: ?(0)
      - **l_tg**: −7.72
      - **l_total**: −23.42
      - **LCOE_pv**: 0.07
- **meta**:
  - **inputs**:
    - **location**:
      - **description**: Selected location
      - **variables**:
        - **latitude**:
          - **description**: Latitude
          - **units**: decimal degree
        - **longitude**:
          - **description**: Longitude
          - **units**: decimal degree
        - **elevation**:
          - **description**: Elevation
          - **units**: m
    - **meteo_data**:
      - **description**: Sources of meteorological data
      - **variables**:

  - **radiation_db**:
    - **description**: Solar radiation database
  - **meteo_db**:
    - **description**: Database used for meteorological variables other than solar radiation
  - **year_min**:
    - **description**: First year of the calculations
  - **year_max**:
    - **description**: Last year of the calculations
  - **use_horizon**:
    - **description**: Include horizon shadows
  - **horizon_db**:
    - **description**: Source of horizon data
- **mounting_system**:
  - **description**: Mounting system
  - **choices**: fixed, vertical_axis, inclined_axis, two_axis
  - **fields**:
    - **slope**:
      - **description**: Inclination angle from the horizontal plane
      - **units**: degree
    - **azimuth**:
      - **description**: Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, −90 = E)
      - **units**: degree
- **pv_module**:
  - **description**: PV module parameters
  - **variables**:
    - **technology**:
      - **description**: PV technology
    - **peak_power**:
      - **description**: Nominal (peak) power of the PV module
      - **units**: kW
    - **system_loss**:
      - **description**: Sum of system losses
      - **units**: %
- **economic_data**:
  - **description**: Economic inputs
  - **variables**:
    - **system_cost**:
      - **description**: Total cost of the PV system
      - **units**: user-defined currency
    - **interest**:
      - **description**: Annual interest
      - **units**: %/y
    - **lifetime**:
      - **description**: Expected lifetime of the PV system
      - **units**: y

- **outputs**:
  - **monthly**:
    - **type**: time series
    - **timestamp**: monthly averages
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
  - **totals**:
    - **type**: time series totals
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **E_y**:
        - **description**: Average annual energy production from the given system
        - **units**: kWh/y
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **H(i)_y**:
        - **description**: Average annual sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/y
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
      - **SD_y**:
        - **description**: Standard deviation of the annual energy production due to year-to-year variation
        - **units**: kWh
      - **l_aoi**:
        - **description**: Angle of incidence loss
        - **units**: %
      - **l_spec**:
        - **description**: Spectral loss
        - **units**: %
      - **l_tg**:
        - **description**: Temperature and irradiance loss
        - **units**: %
      - **l_total**:
        - **description**: Total loss
        - **units**: %
      - **LCOE_pv**:
        - **description**: Levelized cost of the PV electricity
        - **units**: system_cost_currency/kWh

7.2. G3

- **inputs**:
  - **location**:
    - **latitude**: 36.66624290569493
    - **longitude**: −4.461510030795912
    - **elevation**: 4.0
  - **meteo_data**:
    - **radiation_db**: PVGIS-SARAH3
    - **meteo_db**: ERA5
    - **year_min**: 2005
    - **year_max**: 2023
    - **use_horizon**: True
    - **horizon_db**: DEM-calculated
  - **mounting_system**:
    - **fixed**:
      - **slope**:
        - **value**: 26
        - **optimal**: False
      - **azimuth**:
        - **value**: 0
        - **optimal**: False
      - **type**: free-standing
  - **pv_module**:
    - **technology**: c-Si
    - **peak_power**: 6.0
    - **system_loss**: 14.0
  - **economic_data**:
    - **system_cost**: 5994.0
    - **interest**: 5.0
    - **lifetime**: 20
- **outputs**:
  - **monthly**:
    - **fixed**:
      - **month**: 1
      - **E_d**: 21.16
      - **E_m**: 655.88
      - **H(i)_d**: 4.36
      - **H(i)_m**: 135.12
      - **SD_m**: 70.97
      - **month**: 2
      - **E_d**: 23.97
      - **E_m**: 671.19
      - **H(i)_d**: 4.96
      - **H(i)_m**: 138.89
      - **SD_m**: 77.24
      - **month**: 3
      - **E_d**: 26.48
      - **E_m**: 820.85
      - **H(i)_d**: 5.58
      - **H(i)_m**: 172.89
      - **SD_m**: 108.18
      - **month**: 4
      - **E_d**: 29.26
      - **E_m**: 877.71
      - **H(i)_d**: 6.26
      - **H(i)_m**: 187.77
      - **SD_m**: 72.07
      - **month**: 5
      - **E_d**: 31.86
      - **E_m**: 987.76
      - **H(i)_d**: 6.94
      - **H(i)_m**: 215.12
      - **SD_m**: 67.16
      - **month**: 6
      - **E_d**: 33.48
      - **E_m**: 1004.48
      - **H(i)_d**: 7.42
      - **H(i)_m**: 222.69
      - **SD_m**: 32.69
      - **month**: 7
      - **E_d**: 33.57
      - **E_m**: 1040.59
      - **H(i)_d**: 7.57
      - **H(i)_m**: 234.63
      - **SD_m**: 21.18
      - **month**: 8
      - **E_d**: 32.15
      - **E_m**: 996.5
      - **H(i)_d**: 7.23
      - **H(i)_m**: 224.23
      - **SD_m**: 30.23
      - **month**: 9
      - **E_d**: 28.88
      - **E_m**: 866.35
      - **H(i)_d**: 6.37
      - **H(i)_m**: 191.21
      - **SD_m**: 43.8
      - **month**: 10
      - **E_d**: 24.27
      - **E_m**: 752.44
      - **H(i)_d**: 5.26
      - **H(i)_m**: 162.99
      - **SD_m**: 66.87
      - **month**: 11
      - **E_d**: 20.86
      - **E_m**: 625.83
      - **H(i)_d**: 4.37
      - **H(i)_m**: 130.96
      - **SD_m**: 65.09
      - **month**: 12
      - **E_d**: 19.37
      - **E_m**: 600.41
      - **H(i)_d**: 3.98
      - **H(i)_m**: 123.46
      - **SD_m**: 60.4
  - **totals**:
    - **fixed**:
      - **E_d**: 27.12
      - **E_m**: 825.0
      - **E_y**: 9899.99
      - **H(i)_d**: 5.86
      - **H(i)_m**: 178.33
      - **H(i)_y**: 2139.94
      - **SD_m**: 20.73
      - **SD_y**: 248.81
      - **l_aoi**: −2.65
      - **l_spec**: ?(0)
      - **l_tg**: −7.9
      - **l_total**: −22.9
      - **LCOE_pv**: 0.06
- **meta**:
  - **inputs**:
    - **location**:
      - **description**: Selected location
      - **variables**:
        - **latitude**:
          - **description**: Latitude
          - **units**: decimal degree
        - **longitude**:
          - **description**: Longitude
          - **units**: decimal degree
        - **elevation**:
          - **description**: Elevation
          - **units**: m
    - **meteo_data**:
      - **description**: Sources of meteorological data
      - **variables**:
        - **radiation_db**:
          - **description**: Solar radiation database

        - **meteo_db**:
          - **description**: Database used for meteorological variables other than solar radiation
        - **year_min**:
          - **description**: First year of the calculations
        - **year_max**:
          - **description**: Last year of the calculations
        - **use_horizon**:
          - **description**: Include horizon shadows
        - **horizon_db**:
          - **description**: Source of horizon data
  - **mounting_system**:
    - **description**: Mounting system
    - **choices**: fixed, vertical_axis, inclined_axis, two_axis
    - **fields**:
      - **slope**:
        - **description**: Inclination angle from the horizontal plane
        - **units**: degree
      - **azimuth**:
        - **description**: Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, −90 = E)
        - **units**: degree
  - **pv_module**:
    - **description**: PV module parameters
    - **variables**:
      - **technology**:
        - **description**: PV technology
      - **peak_power**:
        - **description**: Nominal (peak) power of the PV module
        - **units**: kW
      - **system_loss**:
        - **description**: Sum of system losses
        - **units**: %
  - **economic_data**:
    - **description**: Economic inputs
    - **variables**:
      - **system_cost**:
        - **description**: Total cost of the PV system
        - **units**: user-defined currency
      - **interest**:
        - **description**: Annual interest
        - **units**: %/y
      - **lifetime**:
        - **description**: Expected lifetime of the PV system
        - **units**: y
- **outputs**:
  - **monthly**:
    - **type**: time series
    - **timestamp**: monthly averages
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
  - **totals**:
    - **type**: time series totals
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **E_y**:
        - **description**: Average annual energy production from the given system
        - **units**: kWh/y
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **H(i)_y**:
        - **description**: Average annual sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/y
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
      - **SD_y**:
        - **description**: Standard deviation of the annual energy production due to year-to-year variation
        - **units**: kWh
      - **l_aoi**:
        - **description**: Angle of incidence loss
        - **units**: %
      - **l_spec**:
        - **description**: Spectral loss
        - **units**: %
      - **l_tg**:
        - **description**: Temperature and irradiance loss
        - **units**: %
      - **l_total**:
        - **description**: Total loss
        - **units**: %
      - **LCOE_pv**:
        - **description**: Levelized cost of the PV electricity
        - **units**: system_cost_currency/kWh

7.3. Total

- **inputs**:
  - **location**:
    - **latitude**: 36.67
    - **longitude**: −4.46
    - **elevation**: 4.0
  - **meteo_data**:
    - **radiation_db**: PVGIS-SARAH3
    - **meteo_db**: ERA5
    - **year_min**: 2005
    - **year_max**: 2023
    - **use_horizon**: True
    - **horizon_db**: DEM-calculated
  - **mounting_system**:
    - **fixed**:
      - **slope**:
        - **value**: 0
        - **optimal**: False
      - **azimuth**:
        - **value**: 0
        - **optimal**: False
      - **type**: free-standing
  - **pv_module**:
    - **technology**: c-Si
    - **peak_power**: 7.62
    - **system_loss**: 14.0
  - **economic_data**:
    - **system_cost**: 1620.0
    - **interest**: 5.0
    - **lifetime**: 20
- **outputs**:
  - **monthly**:
    - **fixed**:
      - **month**: 1
      - **E_d**: 24.66
      - **E_m**: 764.5
      - **H(i)_d**: 4.02
      - **H(i)_m**: 124.73
      - **SD_m**: 80.72
      - **month**: 2
      - **E_d**: 28.64
      - **E_m**: 801.98
      - **H(i)_d**: 4.68
      - **H(i)_m**: 130.98
      - **SD_m**: 89.6
      - **month**: 3
      - **E_d**: 32.49
      - **E_m**: 1007.12
      - **H(i)_d**: 5.39
      - **H(i)_m**: 166.99
      - **SD_m**: 129.64
      - **month**: 4
      - **E_d**: 36.72
      - **E_m**: 1101.39
      - **H(i)_d**: 6.18
      - **H(i)_m**: 185.32
      - **SD_m**: 89.05
      - **month**: 5
      - **E_d**: 40.67
      - **E_m**: 1260.93
      - **H(i)_d**: 6.97
      - **H(i)_m**: 215.93
      - **SD_m**: 85.6
      - **month**: 6
      - **E_d**: 43.1
      - **E_m**: 1293.07
      - **H(i)_d**: 7.51
      - **H(i)_m**: 225.4
      - **SD_m**: 42.14
      - **month**: 7
      - **E_d**: 43.03
      - **E_m**: 1333.85
      - **H(i)_d**: 7.63
      - **H(i)_m**: 236.45
      - **SD_m**: 27.24
      - **month**: 8
      - **E_d**: 40.58
      - **E_m**: 1257.73
      - **H(i)_d**: 7.17
      - **H(i)_m**: 222.51
      - **SD_m**: 37.92
      - **month**: 9
      - **E_d**: 35.69
      - **E_m**: 1070.62
      - **H(i)_d**: 6.19
      - **H(i)_m**: 185.88
      - **SD_m**: 52.73
      - **month**: 10
      - **E_d**: 29.3
      - **E_m**: 908.44
      - **H(i)_d**: 5.0
      - **H(i)_m**: 155.09
      - **SD_m**: 78.22
      - **month**: 11
      - **E_d**: 24.57
      - **E_m**: 737.07
      - **H(i)_d**: 4.07
      - **H(i)_m**: 121.99
      - **SD_m**: 74.68
      - **month**: 12
      - **E_d**: 22.46
      - **E_m**: 696.29
      - **H(i)_d**: 3.66
      - **H(i)_m**: 113.58
      - **SD_m**: 68.14
  - **totals**:
    - **fixed**:
      - **E_d**: 33.51
      - **E_m**: 1019.42
      - **E_y**: 12232.97
      - **H(i)_d**: 5.71
      - **H(i)_m**: 173.74
      - **H(i)_y**: 2084.82
      - **SD_m**: 24.53
      - **SD_y**: 294.44
      - **l_aoi**: −2.83
      - **l_spec**: 0.0
      - **l_tg**: −7.86
      - **l_total**: −23.01
      - **LCOE_pv**: 0.06
- **meta**:
  - **inputs**:
    - **location**:
      - **description**: Selected location
      - **variables**:
        - **latitude**:
          - **description**: Latitude
          - **units**: decimal degree
        - **longitude**:
          - **description**: Longitude
          - **units**: decimal degree
        - **elevation**:
          - **description**: Elevation
          - **units**: m
    - **meteo_data**:
      - **description**: Sources of meteorological data
      - **variables**:
        - **radiation_db**:
          - **description**: Solar radiation database

      - **meteo_db**:
        - **description**: Database used for meteorological variables other than solar radiation
      - **year_min**:
        - **description**: First year of the calculations
      - **year_max**:
        - **description**: Last year of the calculations
      - **use_horizon**:
        - **description**: Include horizon shadows
      - **horizon_db**:
        - **description**: Source of horizon data
  - **mounting_system**:
    - **description**: Mounting system
    - **choices**: fixed, vertical_axis, inclined_axis, two_axis
    - **fields**:
      - **slope**:
        - **description**: Inclination angle from the horizontal plane
        - **units**: degree
      - **azimuth**:
        - **description**: Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, −90 = E)
        - **units**: degree
  - **pv_module**:
    - **description**: PV module parameters
    - **variables**:
      - **technology**:
        - **description**: PV technology
      - **peak_power**:
        - **description**: Nominal (peak) power of the PV module
        - **units**: kW
      - **system_loss**:
        - **description**: Sum of system losses
        - **units**: %
  - **economic_data**:
    - **description**: Economic inputs
    - **variables**:
      - **system_cost**:
        - **description**: Total cost of the PV system
        - **units**: user-defined currency
      - **interest**:
        - **description**: Annual interest
        - **units**: %/y
      - **lifetime**:
        - **description**: Expected lifetime of the PV system
        - **units**: y
- **outputs**:
  - **monthly**:
    - **type**: time series
    - **timestamp**: monthly averages
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
  - **totals**:
    - **type**: time series totals
    - **variables**:
      - **E_d**:
        - **description**: Average daily energy production from the given system
        - **units**: kWh/d
      - **E_m**:
        - **description**: Average monthly energy production from the given system
        - **units**: kWh/mo
      - **E_y**:
        - **description**: Average annual energy production from the given system
        - **units**: kWh/y
      - **H(i)_d**:
        - **description**: Average daily sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/d
      - **H(i)_m**:
        - **description**: Average monthly sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/mo
      - **H(i)_y**:
        - **description**: Average annual sum of global irradiation per square meter received by the modules of the given system
        - **units**: kWh/m2/y
      - **SD_m**:
        - **description**: Standard deviation of the monthly energy production due to year-to-year variation
        - **units**: kWh
      - **SD_y**:
        - **description**: Standard deviation of the annual energy production due to year-to-year variation
        - **units**: kWh
      - **l_aoi**:
        - **description**: Angle of incidence loss
        - **units**: %
      - **l_spec**:
        - **description**: Spectral loss
        - **units**: %
      - **l_tg**:
        - **description**: Temperature and irradiance loss
        - **units**: %
      - **l_total**:
        - **description**: Total loss
        - **units**: %
      - **LCOE_pv**:
        - **description**: Levelized cost of the PV electricity
        - **units**: system_cost_currency/kWh